

Applied Statistics (MATH10096)

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GLM checking and inference

Checking: residuals

- ↪ Model checking for GLMs is performed using residuals, in the same way as for linear models.
- ↪ However, in the case of GLMs standardization of the residuals is particularly necessary.
- ↪ For GLMs, the main reason for not simply examining the raw residuals, $\hat{\epsilon}_i = y_i - \hat{\mu}_i$, is the difficulty of checking the validity of the assumed mean variance relationship from the raw residuals.

GLM checking and inference

Checking: residuals

- ↪ For example, if a Poisson model is used, then the variance of the residuals should increase in direct proportion to the size of the fitted values, $\hat{\mu}_i$.
- ↪ However, if raw residuals are plotted against fitted values, it is very difficult to judge whether the residual variability is increasing in proportion to the mean, as opposed to, say, the square root or square of the mean.
- ↪ For this reason it is usual to standardize GLM residuals so that, if the model assumptions are correct, the standardized residuals should have approximately equal variance and behave, as far as possible, like residuals from a linear model.

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Checking: Pearson residuals

- ↪ The most obvious way to standardize the residuals is to divide them by a quantity proportional to their standard deviation according to the fitted model.
- ↪ This gives rise to the **Pearson residuals**, defined as

$$\hat{\epsilon}_i = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}},$$

which **should approximately have zero mean and variance ϕ** , if the model is correct.

- ↪ These residuals should not display any trend when plotted against the fitted values, or any predictors (whether included in the model or not), if the model is correctly specified.
- ↪ **Further, if the spread of the residuals is not constant across the fitted values, then this suggests that the assumed mean-variance relationship is not correct.**

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Estimating the dispersion parameter ϕ

- As already mentioned, when ϕ is known, the asymptotic distribution of $\hat{\beta}$ can be used directly for finding confidence intervals for β_i s or testing hypotheses about them.
- Remember that, in the large sample limit, the following holds

$$\hat{\beta} \sim N(\beta, (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \phi)$$

- When ϕ is unknown then an estimator is needed and although it is in principle possible to estimate ϕ by maximum likelihood as well, this is rarely done.
- Instead, a possible estimator is based on the sum of squares of the Pearson residuals, resulting in the so-called **Pearson estimator**,

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)},$$

where p is the number of parameters (dimension of β).

GLM checking and inference

Deviance

- The **deviance** is a key concept generalized linear models and plays a similar role to the residual sum of squares in a linear model.
- Intuitively, it measures the deviance of the fitted generalized linear model with respect to a **perfect model** for the sample $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$.
- This perfect model is known as the **saturated model**: the model with one parameter per data point (and thus n parameters) and perfect fit, i.e., $\hat{\mu} = \mathbf{y}$.
- A perfect fit sounds good but the saturated model is uninformative because it does not summarize the data but merely repeats them in full and it thus does not have the advantages that a simpler model has because of its parsimony, such as better estimation of the true relation between the response variable and predictors.
- However, the full model gives a baseline for measuring the discrepancy for an intermediate model with, say, p parameters.

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Deviance

- ↪ Let $l(\hat{\beta})$ and l_S denote the maximized log likelihood for the model under consideration and for the saturated model, respectively.
- ↪ Note that l_S is the highest value that the log likelihood could possibly have, given the data.
- ↪ Formally, the deviance is defined as

$$D = 2\{l_S - l(\hat{\beta})\}$$

- ↪ The log likelihood $l(\hat{\beta})$ is always smaller than l_S .
- ↪ As a consequence, the deviance is always greater than or equal to zero.
- ↪ The greater the deviance, the poorer the fit. Equivalently, deviance decreases as the model fits the data more closely.

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Deviance

- ↪ For the fitted model under consideration, let $\hat{\theta}_i$ denote the estimate of the canonical parameter θ_i , corresponding to the estimated mean $\hat{\mu}_i$.
- ↪ Similarly, let $\tilde{\theta}_i$ denote the estimate of θ_i for the saturated model, with corresponding mean given by $\tilde{\mu}_i = y_i$.
- ↪ Provided that the response observations are independent and for an exponential family distribution, with $a_i(\phi) = \phi/\omega_i$, the deviance D definition simplifies to

$$\begin{aligned} D &= 2 \sum_{i=1}^n \left\{ \frac{\omega_i [y_i \tilde{\theta}_i - b(\tilde{\theta}_i)]}{\phi} + c(y_i, \phi) - \frac{\omega_i [y_i \hat{\theta}_i - b(\hat{\theta}_i)]}{\phi} - c(y_i, \phi) \right\} \phi \\ &= \sum_{i=1}^n 2\omega_i \{y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\}. \end{aligned}$$

- ↪ Notice that the deviance does not depend on ϕ , as it cancels out.

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Deviance: Poisson regression

- ↪ For a Poisson GLM we have that $\hat{\theta}_i = \log(\hat{\mu}_i)$ and $b(\hat{\theta}_i) = e^{\hat{\theta}_i} = \hat{\mu}_i$.
- ↪ Similarly, for the saturated model, for which $\tilde{\mu}_i = y_i$, we have that $\tilde{\theta}_i = \log(y_i)$ and $b(\tilde{\theta}_i) = y_i$.
- ↪ Also, $\omega_i = 1$, for all $i = 1, \dots, n$.
- ↪ The deviance is then equal to

$$\begin{aligned} D &= \sum_{i=1}^n 2\{y_i(\log y_i - \log \hat{\mu}_i) - y_i + \hat{\mu}_i\} \\ &= \sum_{i=1}^n 2 \left\{ y_i \log \left(\frac{y_i}{\hat{\mu}_i} \right) - y_i + \hat{\mu}_i \right\}. \end{aligned}$$

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Deviance: Gaussian GLM

→ For a normal GLM we have that $\hat{\theta}_i = \hat{\mu}_i$ and $b(\hat{\theta}_i) = \frac{\hat{\theta}_i^2}{2} = \frac{\hat{\mu}_i^2}{2}$.

→ Similarly, $\tilde{\theta}_i = y_i$ and $b(\tilde{\theta}_i) = \frac{y_i^2}{2}$ for the saturated model.

→ We further have $\omega_i = 1, \forall i$.

→ So the deviance equals to

$$\begin{aligned} D &= \sum_{i=1}^n 2 \left\{ y_i(y_i - \hat{\mu}_i) - \frac{y_i^2}{2} + \frac{\hat{\mu}_i^2}{2} \right\} \\ &= \sum_{i=1}^n (y_i^2 - 2y_i\hat{\mu}_i + \hat{\mu}_i^2) \\ &= \sum_{i=1}^n (y_i - \hat{\mu}_i)^2. \end{aligned}$$

→ For ordinary linear models this is the residual sum of squares: $\|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 = \|\mathbf{y} - \hat{\boldsymbol{\mu}}\|^2$

GLM checking and inference

Scaled deviance

→ A quantity related to the deviance is the **scaled deviance**, given by

$$D^* = \frac{D}{\phi} = 2\{l_S - l(\hat{\beta})\}.$$

→ If $\phi = 1$, such as for the Poisson distribution, the deviance and the scaled deviance are the same.

→ Remember the following result from likelihood theory (known as **Generalized likelihood ratio test**): when testing the null hypothesis that model M_0 , with p_0 parameters, nested within model M_1 with p_1 parameters, then the following large sample result holds if the null hypothesis is compatible with the data

$$2\{l(\hat{\beta}_1) - l(\hat{\beta}_0)\} \sim \chi^2_{p_1 - p_0}.$$

→ Therefore, we might expect that, if the model under consideration is correct (and assuming it has p parameters), then approximately

$$D^* \sim \chi^2_{n-p},$$

in the large sample limit.

GLM checking and inference

Scaled deviance

- Technically, the limiting argument justifying this large sample approximation is bogus, since it relies on the number of parameters in the model staying fixed, while the sample size tends to infinity, but the saturated model has as many parameters as data points.
- Asymptotic results are available for some of the distributions in the exponential family, to justify this result, as a large sample approximation under many circumstances, and it is exact for the normal case.
- In this case, however, similar to the case of a Gamma or an Inverse Gaussian distribution, we do not know the dispersion parameter ϕ and so this test usually cannot be used directly as a goodness of fit statistic.
- For the binomial and Poisson distributions, $\phi = 1$, and so the test is practical.
- However, the accuracy of the approximation is dubious for small datasets.
- In the case of the binomial distribution with binary data, the approximation completely breaks down.

GLM checking and inference

Scaled deviance

- ↪ If we want to compare a GLM model M_0 , with p_0 predictors, that is nested within a GLM model M_1 , with p_1 predictors (and so $p_0 < p_1$) using the GLRT, then the test statistic can be re-written in terms of D_0^* and D_1^* to obtain the approximate result that under the null hypothesis that the simpler model, M_0 , is compatible with the data:

$$\begin{aligned} D_0^* - D_1^* &= 2\{l_S - l(\hat{\beta}_0)\} - 2\{l_S - l(\hat{\beta}_1)\} \\ &= 2\{l(\hat{\beta}_1) - l(\hat{\beta}_0)\} \sim \chi_{p_1 - p_0}^2. \end{aligned}$$

- ↪ This is **only useful if the scale parameter ϕ is known so that D^* can be calculated.**
- ↪ Note that the scaled deviance D^* apparently removes the effect of ϕ but it is, in fact, still dependent on ϕ , since this is hidden in the likelihood.

GLM checking and inference

Scaled deviance

- Of course, we can use the previous result with $\hat{\phi}$ plugged into D_j^* to obtain an estimate $\hat{D}_j^*$, $j \in \{0, 1\}$, but a slightly better approximation is the one presented now.
- Under H_0 , the two following approximate results hold

$$D_0^* - D_1^* \sim \chi_{p_1 - p_0}^2 \quad \text{and} \quad D_1^* \sim \chi_{n - p_1}^2,$$

and, if $D_0^* - D_1^*$ and D_1^* are treated as asymptotically independent, this implies that

$$\begin{aligned} F &= \frac{(D_0^* - D_1^*) / (p_1 - p_0)}{D_1^* / (n - p_1)} \\ &= \frac{(D_0 - D_1) / \phi / (p_1 - p_0)}{D_1 / \phi / (n - p_1)} \\ &= \frac{(D_0 - D_1) / (p_1 - p_0)}{D_1 / (n - p_1)} \sim F_{p_1 - p_0, n - p_1}, \end{aligned}$$

in the large sample limit.

- Of course, this result is exactly true in the ordinary linear model special case (i.e., for a Gaussian response and the identity link function).

GLM checking and inference

Deviance residuals

- ↪ The deviance can also provide alternative scaled residuals to the Pearson residuals.
- ↪ Indeed, and in practice, the distribution of the Pearson residuals can be quite asymmetric around zero, so that their behavior is not as close to ordinary linear model residuals as might be hoped for.
- ↪ The **deviance residuals** are often preferable in this respect.
- ↪ The deviance residuals are the generalization of the residuals $\hat{\epsilon}_i = y_i - \hat{\mu}_i$ from the ordinary linear model.
- ↪ They are constructed using the analogy that for ordinary linear models, $D = \sum_{i=1}^n (y_i - \hat{\mu}_i)^2 = \sum_{i=1}^n \hat{\epsilon}_i^2$ is made up of the sum of the squared residuals.
- ↪ That is, the residuals are the square roots of the components of the deviance with the appropriate sign attached.

GLM checking and inference

Deviance residuals

↪ So, writing d_i as the component of the deviance contributed by the i th observation we have

$$D = \sum_{i=1}^n d_i,$$

and, by analogy with the ordinary linear model, we can define

$$\hat{\epsilon}_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}.$$

↪ As required, the sum of squares of these 'deviance residuals' gives the deviance itself.

GLM checking and inference

Deviance residuals

- ↪ This definition has interesting distributional consequences. It can be shown that $\hat{\epsilon}_i^D$ follow, approximately, a $N(0, \phi)$ distribution if the model holds.
- ↪ The approximate normality of the deviance residuals allows for evaluating how well the assumption of the response distribution is satisfied.
- ↪ The speed of this asymptotic convergence and the effective validity of the approximation itself largely depends on several aspects: distribution of the response, sample size, etc.
- ↪ It is however possible to have severe departures from normality even if the model is perfectly correct.